

# Augmentations and basic CNNs

Mark Blumenau

# Augmentations

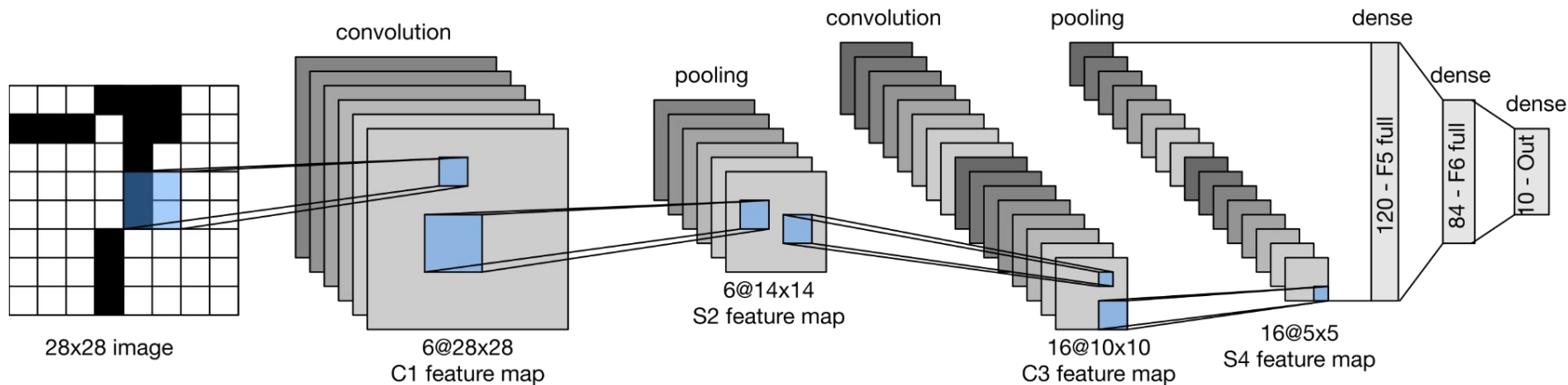
- 1) “Increase” the dataset size for free
- 2) “Regularize” the model

Apply to the current batch, or generate in advance (depending on what you don't mind losing)



# LeNet-5 (1995)

60K params + Augs = less than 1% errors on MNIST



# Mememes about Yann LeCun



**Yann LeCun**

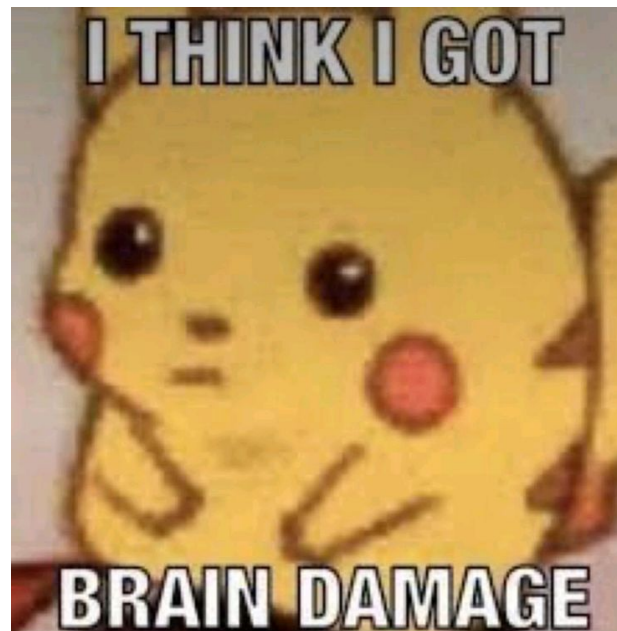
@ylecun

Training with large minibatches is bad for your health.  
More importantly, it's bad for your test error.  
Friends dont let friends use minibatches larger than 32.

[arxiv.org/abs/1804.07612](https://arxiv.org/abs/1804.07612)

2:00 PM · Apr 26, 2018 · [Facebook](#)

Literally me:



# Is there anything other than MNIST? CIFAR-10

60k images, size 32x32

10 classes

<https://www.cs.toronto.edu/~kriz/learning-features-2009-T.R.pdf>





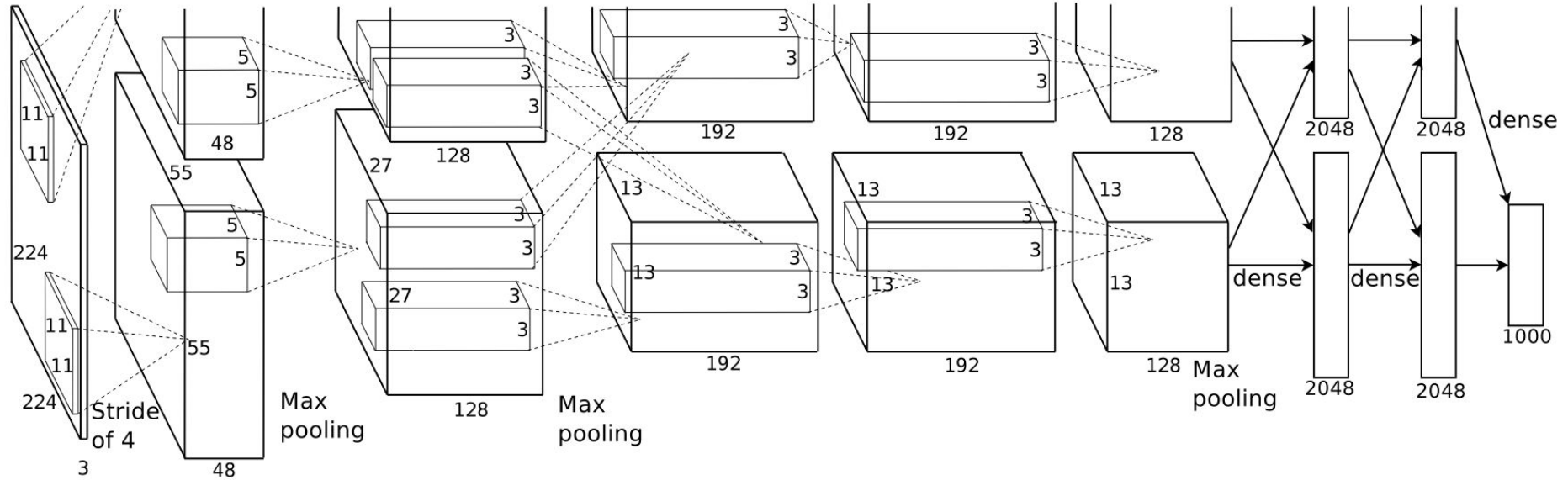
ImageNet

## **ImageNet Large Scale Visual Recognition Challenge (ILSVRC)**

Initially 1 million pictures, 1000 classes

Now 14 mil.

# AlexNet (2012)



[https://proceedings.neurips.cc/paper\\_files/paper/2012/file/c399862d3b9d6b76c8436e924a68c45b-Paper.pdf](https://proceedings.neurips.cc/paper_files/paper/2012/file/c399862d3b9d6b76c8436e924a68c45b-Paper.pdf)

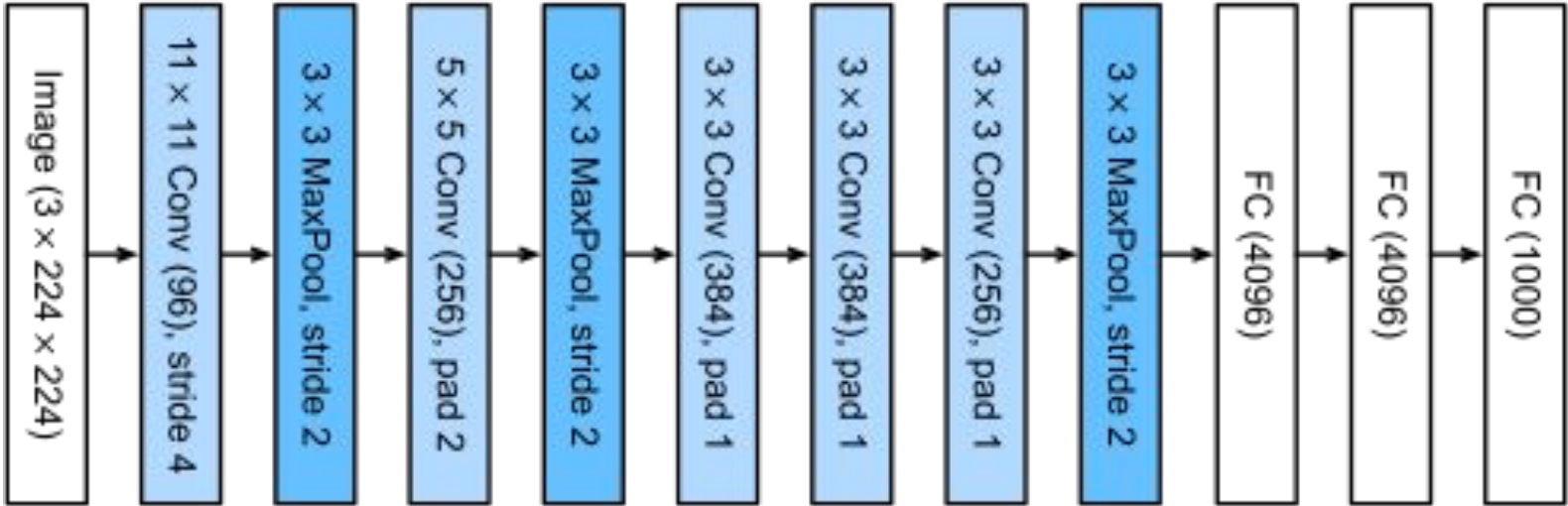


# What about the architecture split?

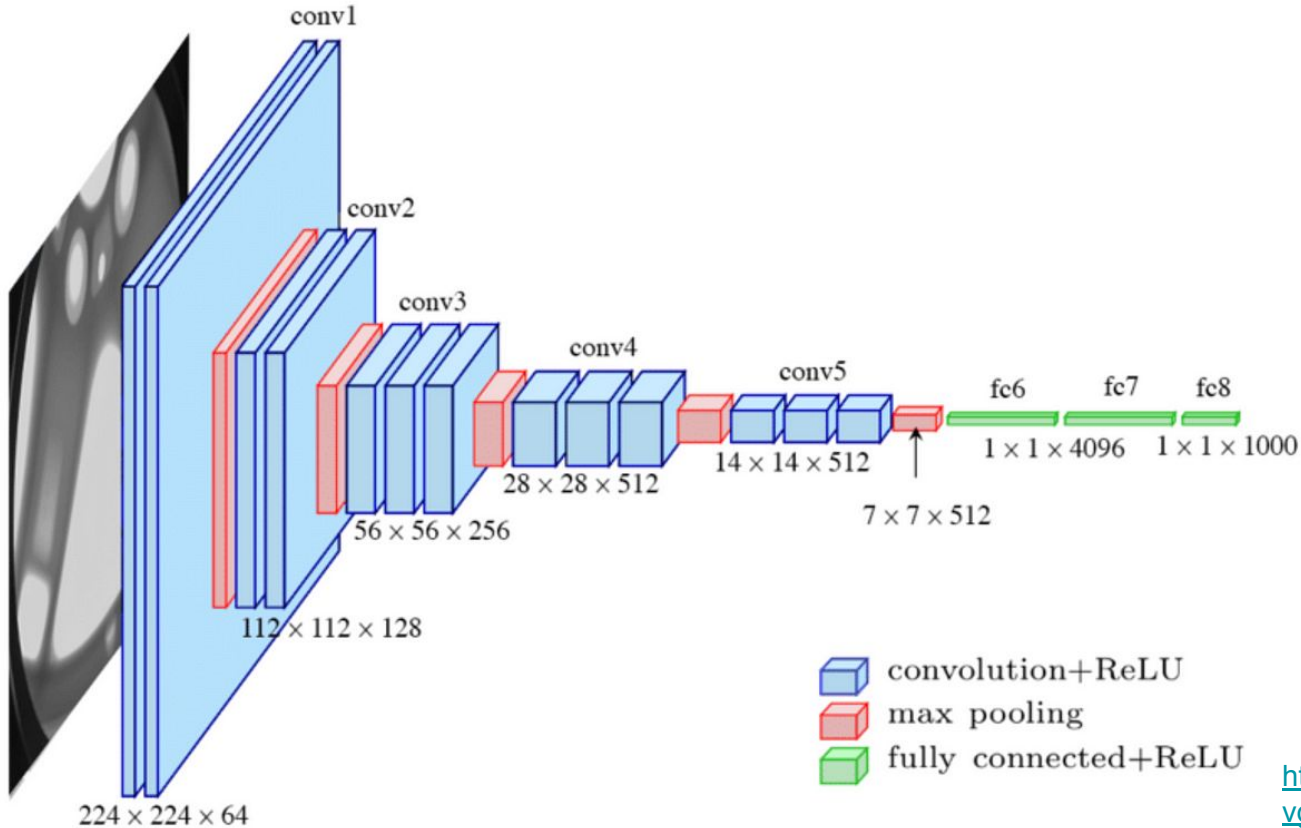
Figure 2: An illustration of the architecture of our CNN, explicitly showing the delineation of responsibilities between the two GPUs. One GPU runs the layer-parts at the top of the figure while the other runs the layer-parts at the bottom. The GPUs communicate only at certain layers. The network's input is 150,528-dimensional, and the number of neurons in the network's remaining layers is given by 253,440–186,624–64,896–64,896–43,264–4096–4096–1000.



A picture that one can understand



# VGG (2014)



<https://lekhuyen.medium.com/an-overview-of-vgg16-and-nin-models-96e4bf398484>

# VGG (2014)

Trick: for bigger NNs, we don't train from scratch, but take weights from small ones.

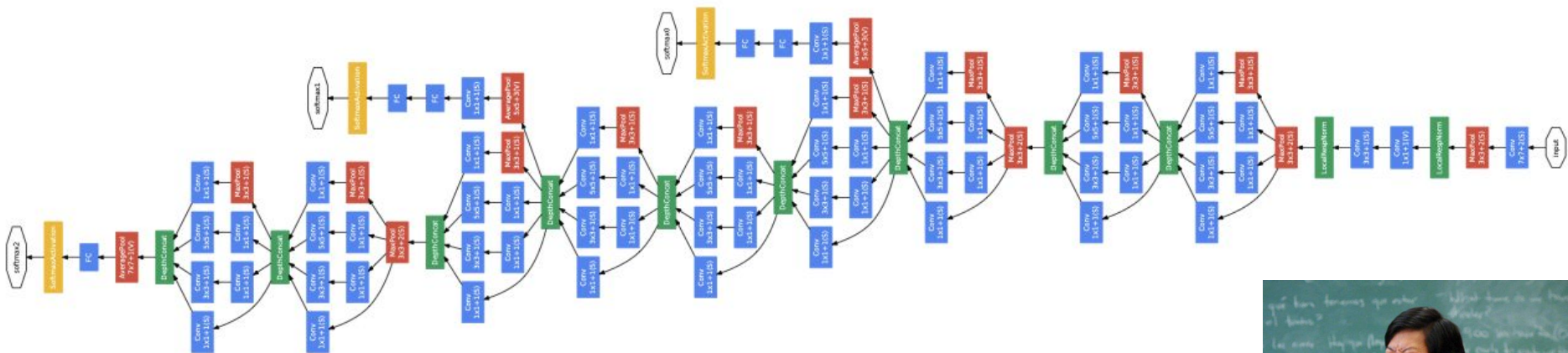
All convolutions are small, there are no large filters.

| ConvNet Configuration       |                        |                               |                                            |                                            |                                                         |
|-----------------------------|------------------------|-------------------------------|--------------------------------------------|--------------------------------------------|---------------------------------------------------------|
| A                           | A-LRN                  | B                             | C                                          | D                                          | E                                                       |
| 11 weight layers            | 11 weight layers       | 13 weight layers              | 16 weight layers                           | 16 weight layers                           | 19 weight layers                                        |
| input (224 × 224 RGB image) |                        |                               |                                            |                                            |                                                         |
| conv3-64                    | conv3-64<br><b>LRN</b> | conv3-64<br><b>conv3-64</b>   | conv3-64<br>conv3-64                       | conv3-64<br>conv3-64                       | conv3-64<br>conv3-64                                    |
| maxpool                     |                        |                               |                                            |                                            |                                                         |
| conv3-128                   | conv3-128              | conv3-128<br><b>conv3-128</b> | conv3-128<br>conv3-128                     | conv3-128<br>conv3-128                     | conv3-128<br>conv3-128                                  |
| maxpool                     |                        |                               |                                            |                                            |                                                         |
| conv3-256<br>conv3-256      | conv3-256<br>conv3-256 | conv3-256<br>conv3-256        | conv3-256<br>conv3-256<br><b>conv1-256</b> | conv3-256<br>conv3-256<br><b>conv3-256</b> | conv3-256<br>conv3-256<br>conv3-256<br><b>conv3-256</b> |
| maxpool                     |                        |                               |                                            |                                            |                                                         |
| conv3-512<br>conv3-512      | conv3-512<br>conv3-512 | conv3-512<br>conv3-512        | conv3-512<br>conv3-512<br><b>conv1-512</b> | conv3-512<br>conv3-512<br><b>conv3-512</b> | conv3-512<br>conv3-512<br>conv3-512<br><b>conv3-512</b> |
| maxpool                     |                        |                               |                                            |                                            |                                                         |
| conv3-512<br>conv3-512      | conv3-512<br>conv3-512 | conv3-512<br>conv3-512        | conv3-512<br>conv3-512<br><b>conv1-512</b> | conv3-512<br>conv3-512<br><b>conv3-512</b> | conv3-512<br>conv3-512<br>conv3-512<br><b>conv3-512</b> |
| maxpool                     |                        |                               |                                            |                                            |                                                         |
| FC-4096                     |                        |                               |                                            |                                            |                                                         |
| FC-4096                     |                        |                               |                                            |                                            |                                                         |
| FC-1000                     |                        |                               |                                            |                                            |                                                         |
| soft-max                    |                        |                               |                                            |                                            |                                                         |

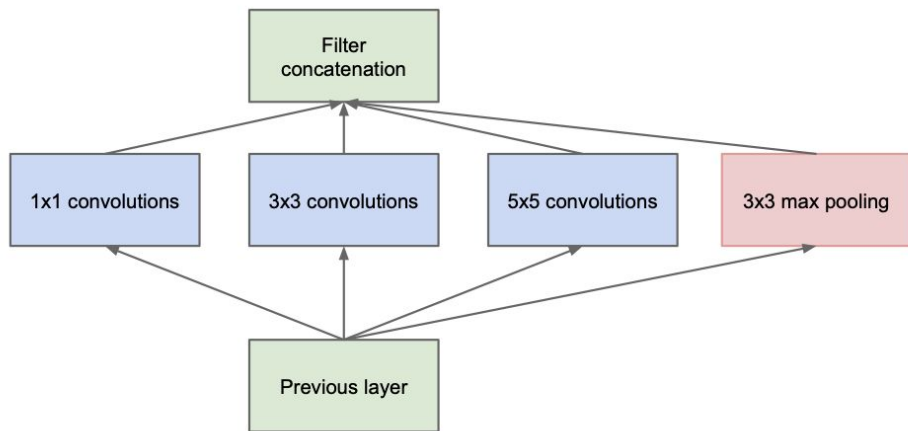
Table 2: **Number of parameters** (in millions).

| Network              | A,A-LRN | B   | C   | D   | E   |
|----------------------|---------|-----|-----|-----|-----|
| Number of parameters | 133     | 133 | 134 | 138 | 144 |

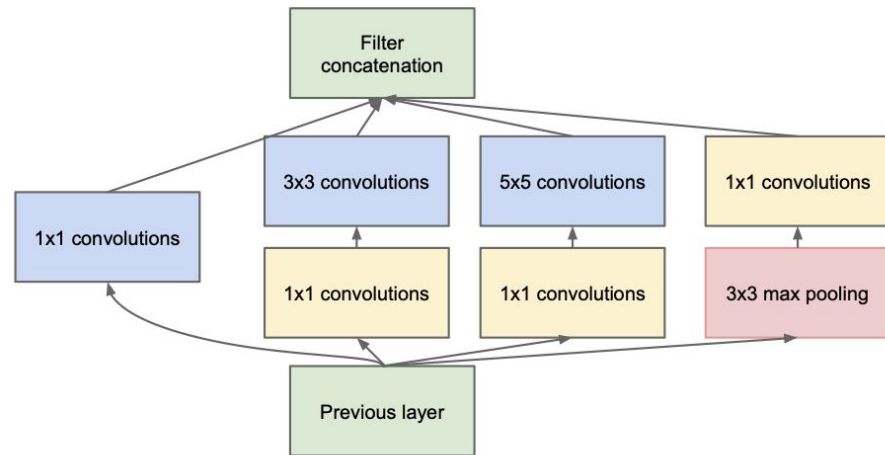
# GoogLeNet aka Going deeper with convolutions (2014)



# Inception module



(a) Inception module, naïve version



(b) Inception module with dimension reductions

# ResNet (2015)

We have a problem

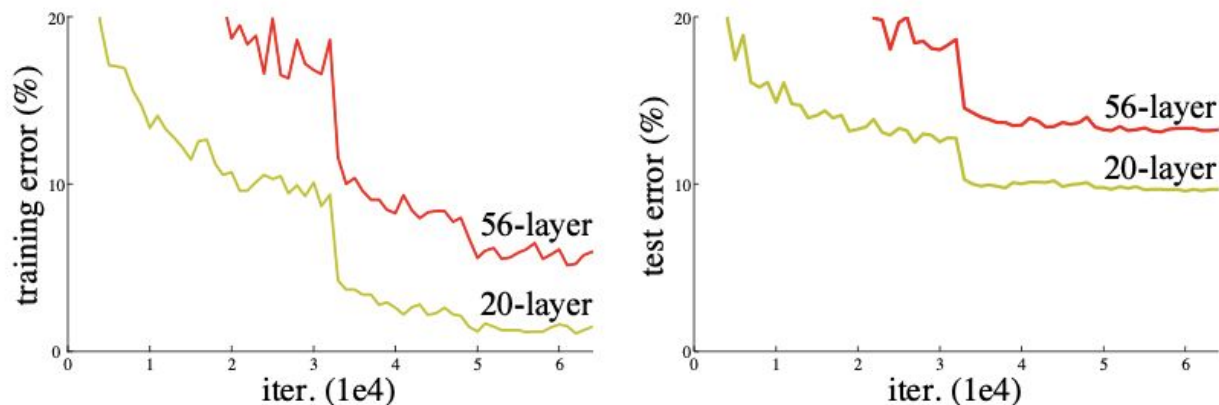
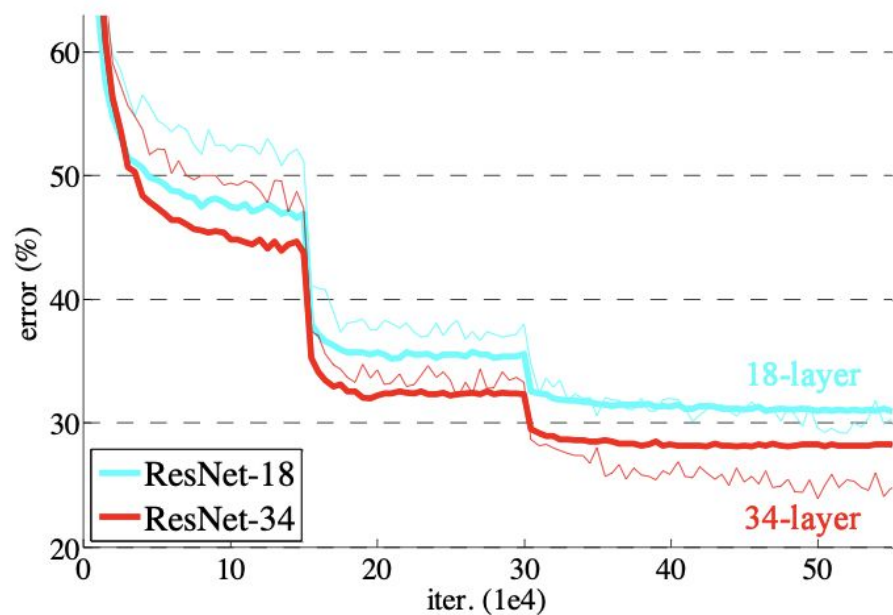
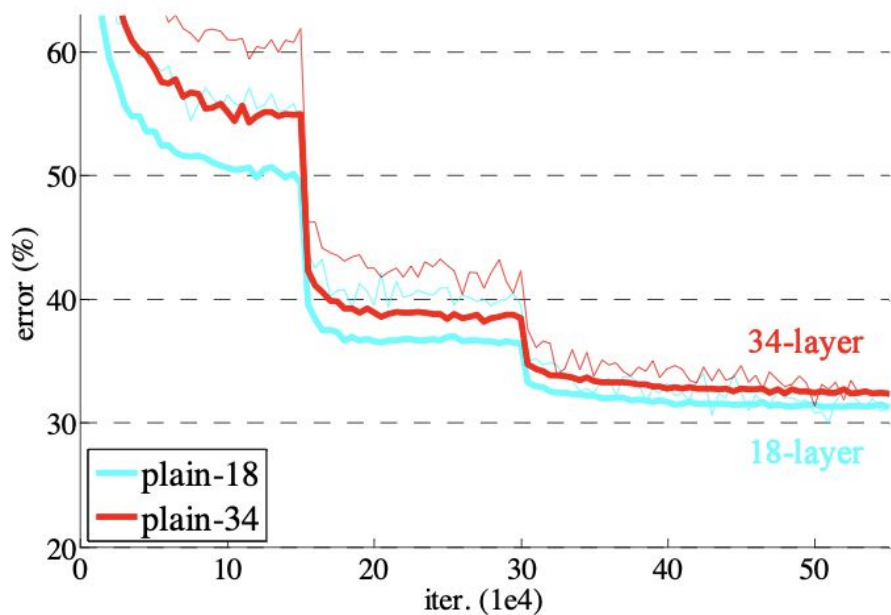


Figure 1. Training error (left) and test error (right) on CIFAR-10 with 20-layer and 56-layer “plain” networks. The deeper network has higher training error, and thus test error. Similar phenomena on ImageNet is presented in Fig. 4.

# Oh, we solved it





How? (Or all we do in CV)

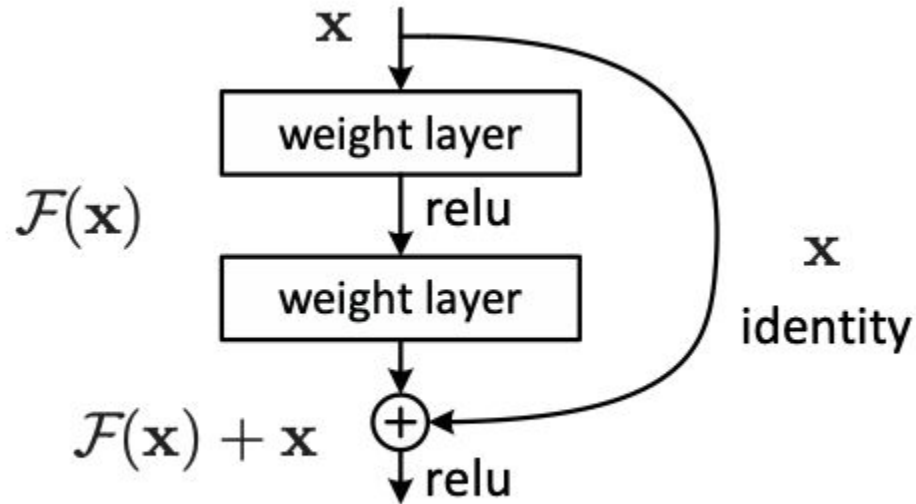
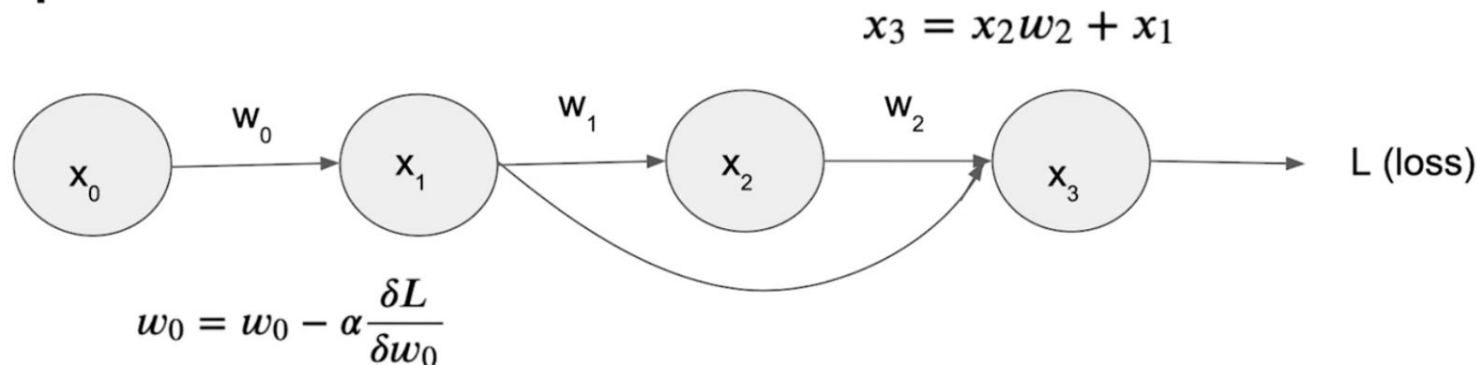


Figure 2. Residual learning: a building block.

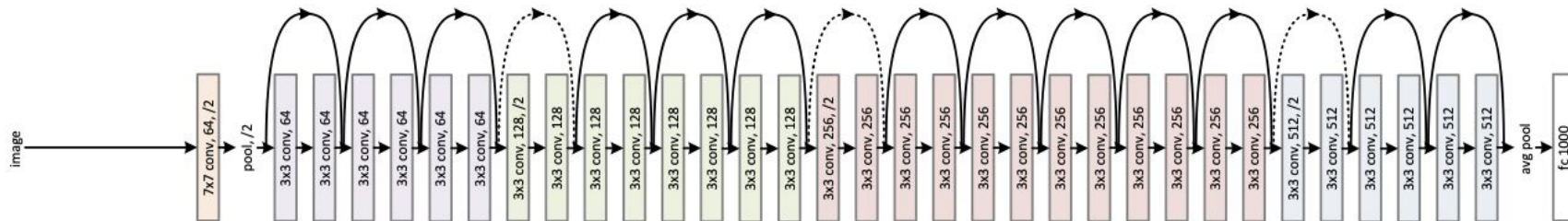
# Skip Connection



$$\begin{aligned} \frac{\delta L}{\delta w_0} &= \frac{\delta L}{\delta x_3} \left( \left( \frac{\delta x_3}{\delta x_2} \frac{\delta x_2}{\delta x_1} \frac{\delta x_1}{\delta w_0} \right) + \left( \frac{\delta x_3}{\delta x_1} \frac{\delta x_1}{\delta w_0} \right) \right) = \\ &= \frac{\delta L}{\delta x_3} \frac{\delta x_1}{\delta w_0} \left( \frac{\delta x_3}{\delta x_2} \frac{\delta x_2}{\delta x_1} + \frac{\delta x_3}{\delta x_1} \right) = \frac{\delta L}{\delta x_3} \frac{\delta x_1}{\delta w_0} \left( \frac{\delta x_3}{\delta x_2} \frac{\delta x_2}{\delta x_1} + 1 \right) \end{aligned}$$

# How does the NN look like

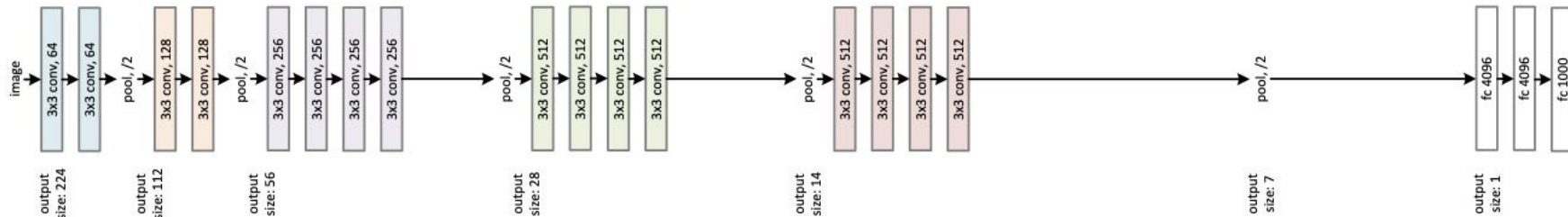
34-layer residual



34-layer plain

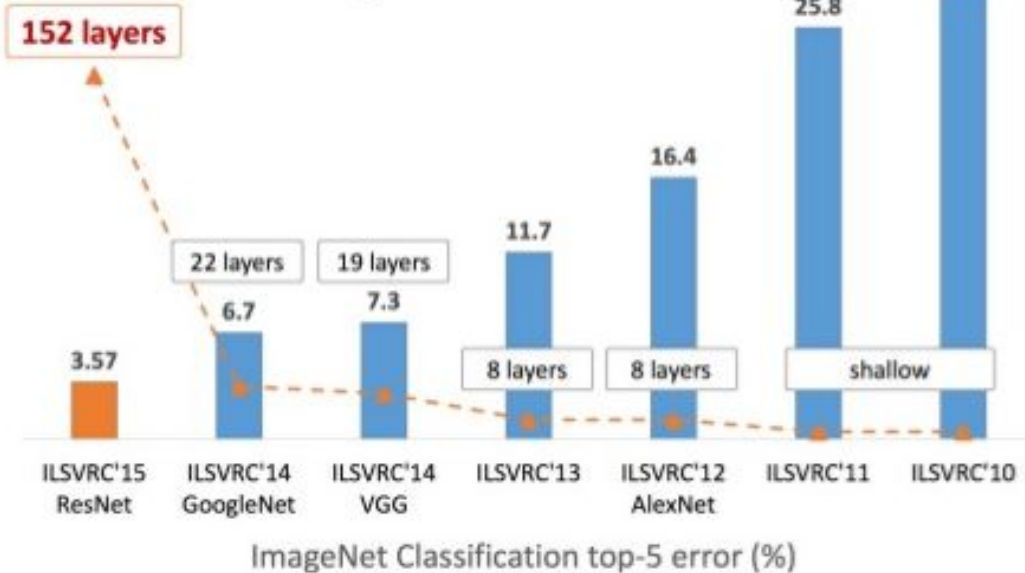


VGG-19



# What we know now

## Revolution of Depth



[https://www.researchgate.net/figure/Performance-of-different-approaches-in-ImageNet-2015-competition\\_fig2\\_309392322](https://www.researchgate.net/figure/Performance-of-different-approaches-in-ImageNet-2015-competition_fig2_309392322)

# What else do we have? (CV course?)

Squeeze-and-Excite

MBConv (see MobileNet/EfficientNet)

Fused MBConv (see EfficientNetV2)

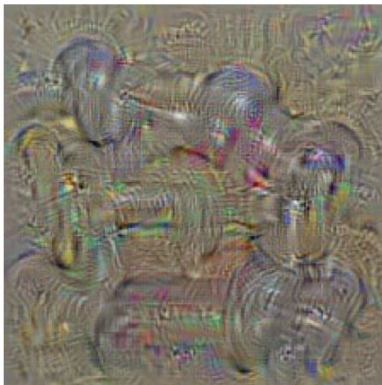
Pyramids!

ConvNext

ViT

...

# xAI



**dumbbell**



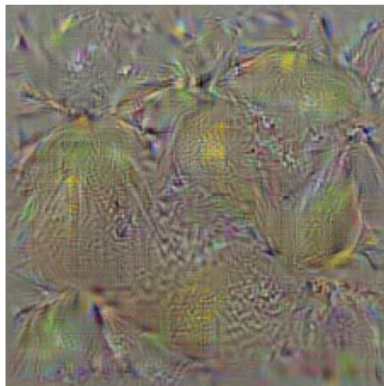
**cup**



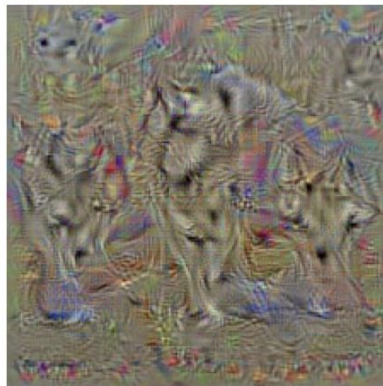
**dalmatian**



**bell pepper**

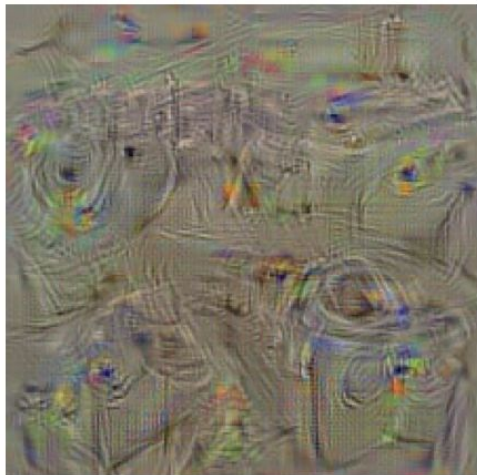


**lemon**

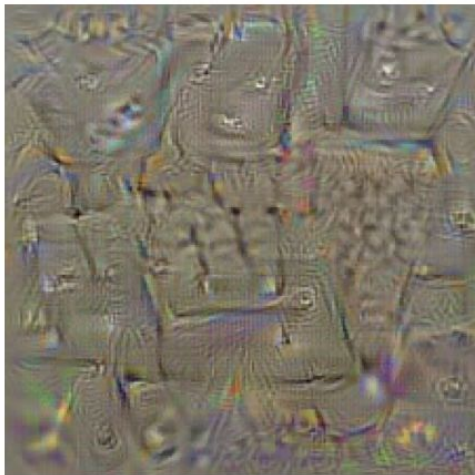


**husky**

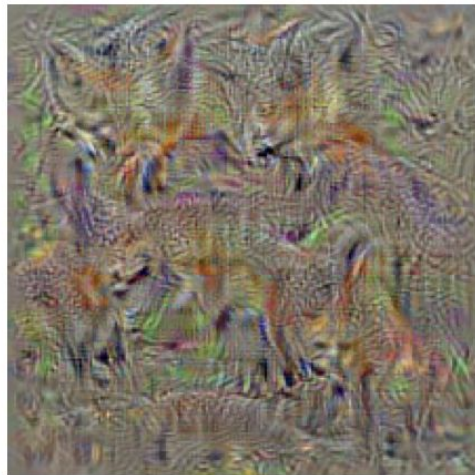




**washing machine**



**computer keyboard**



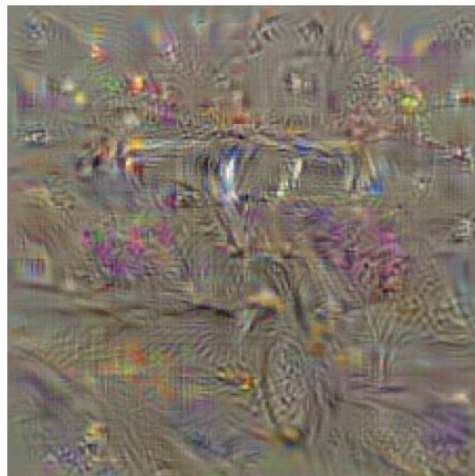
**kit fox**



**goose**

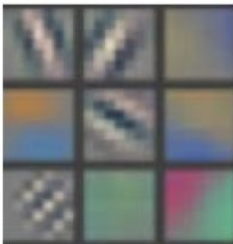


**ostrich**

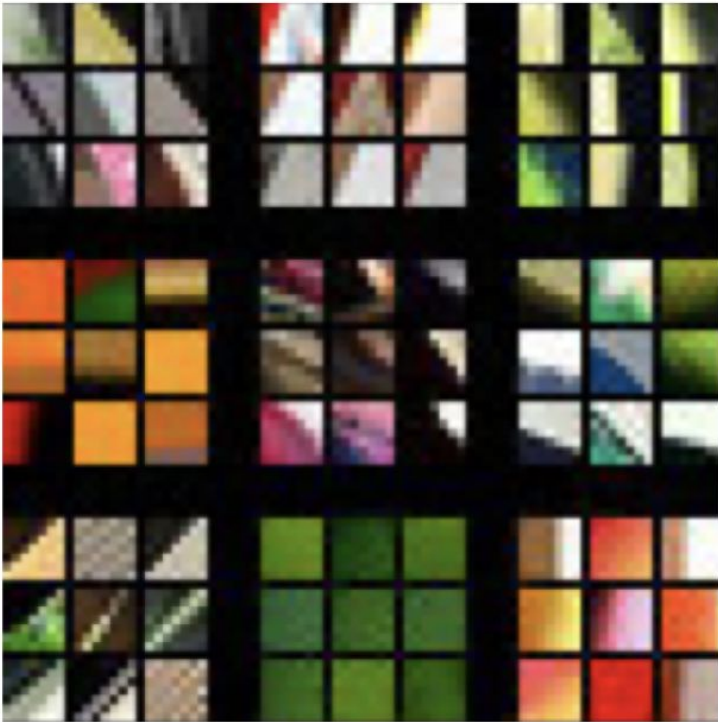


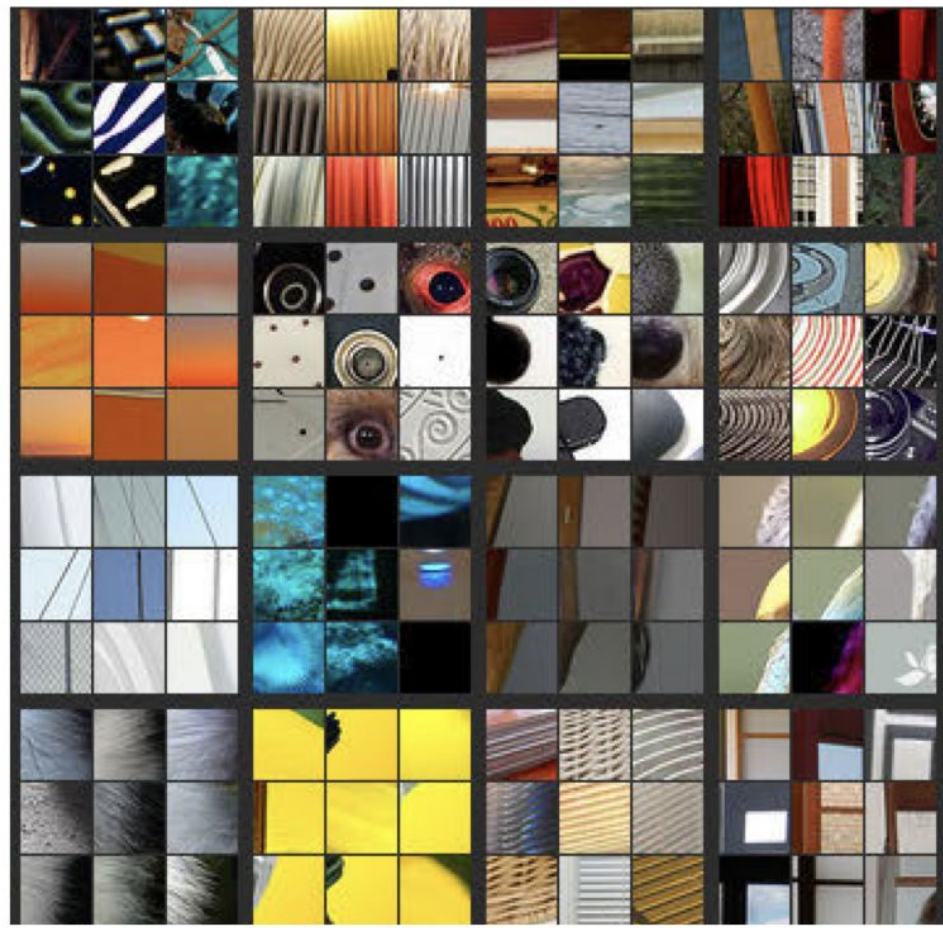
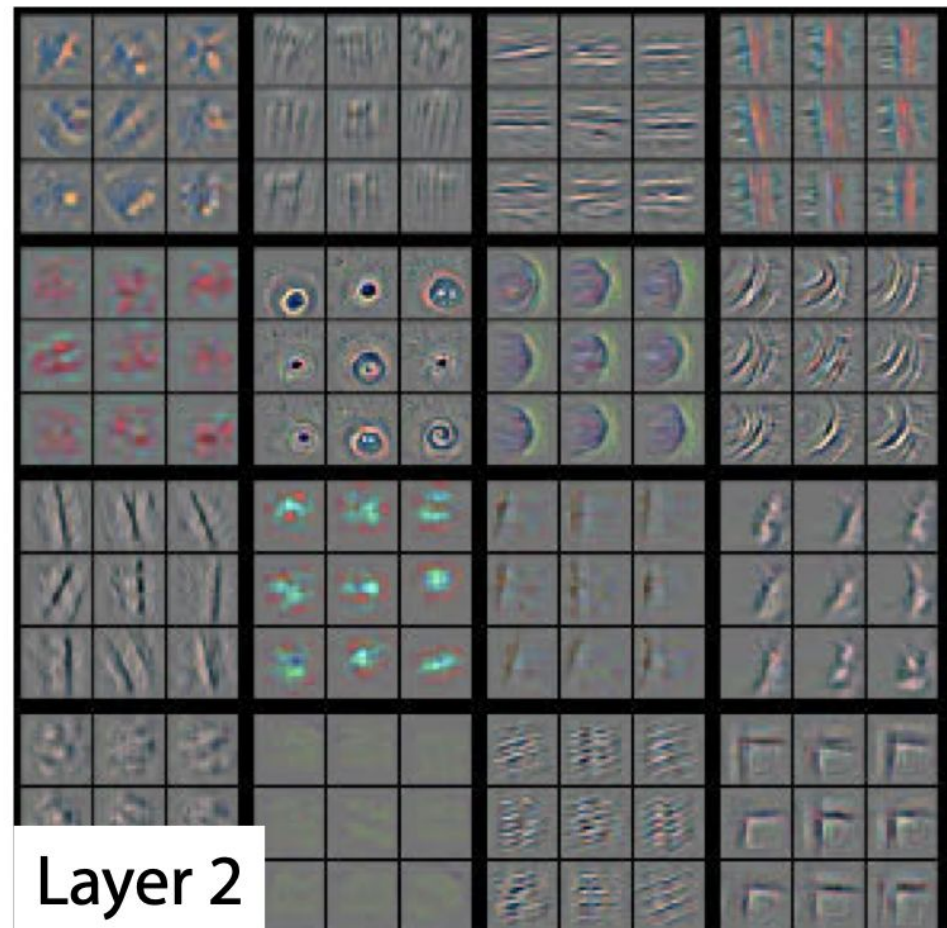
**limousine**



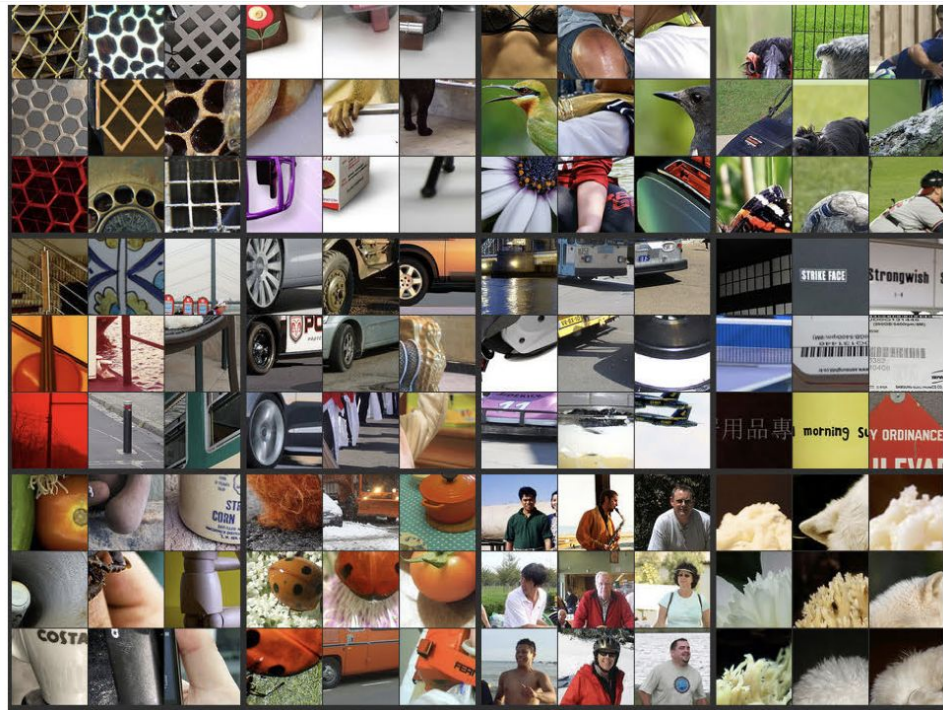
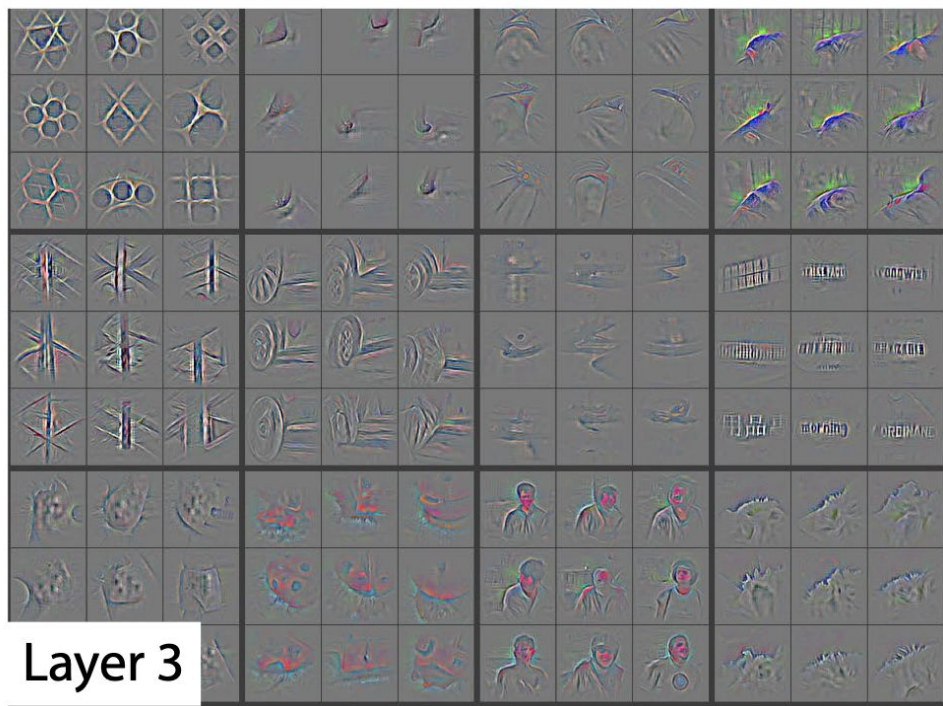


# Layer 1





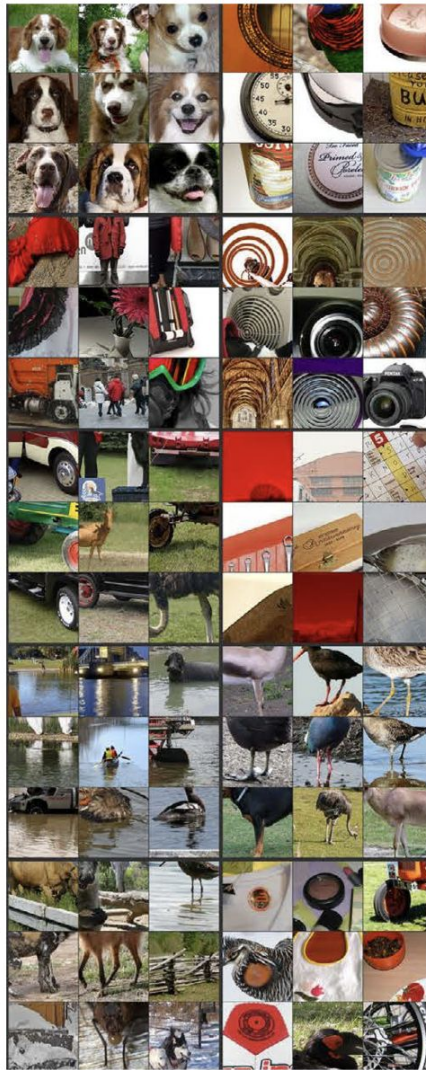






<http://arxiv.org/pdf/1311.2901>

Layer 4



Layer 5

